

Setup Guide OpenVPN Master branch

Version 0.3

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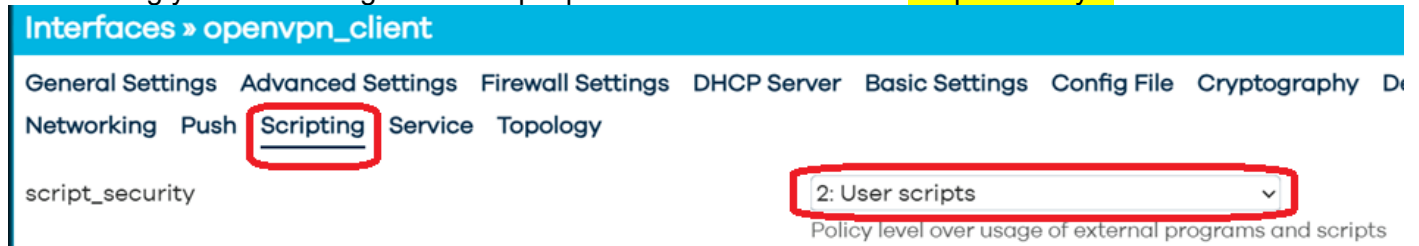
Introduction

On master branch *luci-app-openvpn* is replaced by *luci-proto-openvpn*.

There is no longer a separate entry in the GUI for OpenVPN but it has become a protocol so it is integrated in the interface.

This guide and also the OpenVPN implementation is very much a Work in Progress and subject to change.

When using your own config file to setup OpenVPN make sure to set **script-security 2**



This will allow you to setup the VPN tunnel to you own liking

```
/var/run/openvpn.openvpn_client.conf:
cd /etc/openvpn/openvpn_client
status /var/run/openvpn.openvpn_client.status
syslog openvpn_openvpn_client
tmp-dir /tmp
script-security 2
dev tun1
dev-type tun
mtu-disc no
config '/etc/openvpn/openvpn_client/openvpn_client_config.cfg'
```

Setup Interface

To setup OpenVPN (whether client or server)

Network > Interfaces > Add new interface

Name: choose a name not more than 15 characters, use *openvpn_client*

Protocol: OpenVPN.

Add new interface...

Name	openvpn_client
Protocol	OpenVPN

Click:

Create interface

Now you can setup the OpenVPN interface

Interfaces » openvpn_client

General Settings Advanced Settings Firewall Settings DHCP Server Basic Settings Config File Cryptography Devices Keygen Logging Management

Networking Push Scripting Service Topology

Status	Device: openvpn-openvpn_client Carrier: Absent RX: 0 B (0 Pkts.) TX: 0 B (0 Pkts.)
Protocol	OpenVPN
Disable this interface	☐
Bring up on boot	☒
Almost nothing here prevents you from selecting invalid configuration options which prevent openvpn from starting. Read the manual. Options marked with ¹ are deprecated and will be removed. Options marked with ² are OpenSSL only.	
port	1194 TCP/UDP port # for both local and remote

There are two ways to setup.

The easiest way is to import your config file into the interface.

Click: Config File

Interfaces » openvpn_client

General Settings Advanced Settings Firewall Settings DHCP Server Basic Settings **Config File** Cryptography Device

Networking Push Scripting Service Topology

Drag en drop your OpenVPN config file in the box

Interfaces » openvpn_client

General Settings Advanced Settings Firewall Settings DHCP Server Basic Settings Config File Cryptography Devices Keygen Logging Management
Networking Push Scripting Service Topology

Clear

Clear

Drag and drop an ovpn config file here

Raw OVPN config

Dismiss

Save

Config files can contain all necessary info *inline* including keys certificates and username and password as shown below

Interfaces » openvpn_client

General Settings Advanced Settings Firewall Settings DHCP Server Basic Settings Config File
Networking Push Scripting Service Topology

Clear

Clear

Raw OVPN config

```
remote fr-rbx.vpnunlimitedapp.com
proto udp
port 1197

<auth-user-pass>
REDACTED
</auth-user-pass>

client
dev tun1
reneg-sec 0
persist-tun
ping 5
nobind
allow-compression no
remote-cert-tls server
auth-nocache
data-ciphers CHACHA20-POLY1305:AES-256-GCM:AES-128-GCM
auth sha512

pull-filter ignore "redirect-gateway def1"

#for hotplug scripts use 3:
script-security 3

|ca>
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
</ca>

<cert>
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
</cert>

<key>
-----BEGIN PRIVATE KEY-----
redacted
-----END PRIVATE KEY-----
</key>
```

Often a config file just has `dev tun` so without a number, in this case OpenVPN itself will use the first available number which is **0** so the device will be `tun0`

I think this is bad practice and encourage you to choose a number for the device.

I use `tun1` for my client and `tun2` for my server.

As is chosen to setup the tunnel with a config file with `script-security 2` there is no information about the used `if_device` (the `tun1` device in this example) to make sure applications which use the interface can

extract the I3_device from the config (e.g. for Policy Based Routing) also add the device under Devices > dev:



Interfaces > openvpn_client

General Settings Advanced Settings Firewall Settings DHCP Server Basic Settings Config File Cryptography **Devices**

Networking Push Scripting Service Topology

dev tun/tap device

this way PBR wil now the device used by the openvpn_client interface

```
/etc/config/network/  
config interface 'openvpn_client'  
    option proto 'openvpn'  
    option dev_type 'tun'  
    option mtu_disc 'no'  
    option multipath 'off'  
    option config '/etc/openvpn/openvpn_client/openvpn_client_config.cfg'  
    option script_security '2'  
    option dev 'tun1'
```

Setup Firewall

OpenVPN Client

Create a new firewall zone:

Network > Firewall > Add

Name: vpn_client

Input: reject

Output: accept

Intra zone forward: reject

IPv4 Masquerading: Enabled (ticked)

MSS clamping: Enabled (ticked)

Covered Networks: none

Allow forward from source zones: lan

Firewall - Zone Settings

General Settings Advanced Settings Contrack Settings

This section defines common properties of "vpn_client". The *input* and *output* options set the default policies for the *forward* option describes the policy for forwarded traffic between different networks within the zone. *Covered networks* members of this zone.

Name	vpn_client
Input	reject
Output	accept
Intra zone forward	reject
IPv4 Masquerading	☒ Enable network address and port translation IPv4 typically enabled on the <i>wan</i> zone.
MSS clamping	☒
Covered networks	unspecified

The options below control the forwarding policies between this zone (vpn_client) and other zones. *Destination zones* match forwarded traffic from other zones **targeted at vpn_client**. The forwarding rule is *not* imply a permission to forward from wan to lan as well.

Allow forward to <i>destination zones</i> :	unspecified
Allow forward from <i>source zones</i> :	lan lan: 

In this example I have adapted my config file to use: dev tun1

As we have used script-security 2 and use our own config there is no device information available for the firewall to use so we have to add the device on the Advanced Settings tab under Covered devices:

Firewall - Zone Settings

General Settings Advanced Settings Conntrack Settings

The options below control the forwarding policies between this zone (vpn_client) and other zones. *Destination zones* **vpn_client**. *Source zones* match forwarded traffic from other zones **targeted at vpn_client**. The forwarding rule is *unic* *not* imply a permission to forward from wan to lan as well.

Covered devices

Use this option to classify zone traffic by raw, non-uci

Alternatively you can add an extra interface with name: *tun1_client*, protocol: *unmanaged* and device: *tun1*

You can the use this interface *tun1_client* to add under Covered networks instead of adding the device under Covered devices

OpenVPN Server

Open up firewall for the server port

Allow Input, Output and Intra zone forward

Allow from VPN to LAN , from VPN to WAN if you want internet for the connected clients and Allow from LAN to VPN if you want bidirectional traffic